

UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS  
General Certificate of Education Advanced Subsidiary Level and Advanced Level  
Advanced International Certificate of Education

**MATHEMATICS**

**9709/06**

**STATISTICS**

**0390/06**

Paper 6 Probability & Statistics 1 **(S1)**

October/November 2005

**1 hour 15 minutes**

Additional materials: Answer Booklet/Paper  
Graph paper  
List of Formulae (MF9)

**READ THESE INSTRUCTIONS FIRST**

If you have been given an Answer Booklet, follow the instructions on the front cover of the Booklet.  
Write your Centre number, candidate number and name on all the work you hand in.  
Write in dark blue or black pen on both sides of the paper.  
You may use a soft pencil for any diagrams or graphs.  
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.  
Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.  
At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [ ] at the end of each question or part question.  
The total number of marks for this paper is 50.  
Questions carrying smaller numbers of marks are printed earlier in the paper, and questions carrying larger numbers of marks later in the paper.  
The use of an electronic calculator is expected, where appropriate.  
You are reminded of the need for clear presentation in your answers.

This document consists of **3** printed pages and **1** blank page.



- 1 A study of the ages of car drivers in a certain country produced the results shown in the table.

**Percentage of drivers in each age group**

|         | Young | Middle-aged | Elderly |
|---------|-------|-------------|---------|
| Males   | 40    | 35          | 25      |
| Females | 20    | 70          | 10      |

Illustrate these results diagrammatically. [4]

- 2 Boxes of sweets contain toffees and chocolates. Box *A* contains 6 toffees and 4 chocolates, box *B* contains 5 toffees and 3 chocolates, and box *C* contains 3 toffees and 7 chocolates. One of the boxes is chosen at random and two sweets are taken out, one after the other, and eaten.

(i) Find the probability that they are both toffees. [3]

(ii) Given that they are both toffees, find the probability that they both came from box *A*. [3]

- 3 A staff car park at a school has 13 parking spaces in a row. There are 9 cars to be parked.

(i) How many different arrangements are there for parking the 9 cars and leaving 4 empty spaces? [2]

(ii) How many different arrangements are there if the 4 empty spaces are next to each other? [3]

(iii) If the parking is random, find the probability that there will **not** be 4 empty spaces next to each other. [2]

- 4 A group of 10 married couples and 3 single men found that the mean age  $\bar{x}_w$  of the 10 women was 41.2 years and the standard deviation of the women's ages was 15.1 years. For the 13 men, the mean age  $\bar{x}_m$  was 46.3 years and the standard deviation was 12.7 years.

(i) Find the mean age of the whole group of 23 people. [2]

(ii) The individual women's ages are denoted by  $x_w$  and the individual men's ages by  $x_m$ . By first finding  $\Sigma x_w^2$  and  $\Sigma x_m^2$ , find the standard deviation for the whole group. [5]

- 5 A box contains 300 discs of different colours. There are 100 pink discs, 100 blue discs and 100 orange discs. The discs of each colour are numbered from 0 to 99. Five discs are selected at random, one at a time, with replacement. Find

(i) the probability that no orange discs are selected, [1]

(ii) the probability that exactly 2 discs with numbers ending in a 6 are selected, [3]

(iii) the probability that exactly 2 orange discs with numbers ending in a 6 are selected, [2]

(iv) the mean and variance of the number of pink discs selected. [2]

- 6** In a competition, people pay \$1 to throw a ball at a target. If they hit the target on the first throw they receive \$5. If they hit it on the second or third throw they receive \$3, and if they hit it on the fourth or fifth throw they receive \$1. People stop throwing after the first hit, or after 5 throws if no hit is made. Mario has a constant probability of  $\frac{1}{5}$  of hitting the target on any throw, independently of the results of other throws.
- (i)** Mario misses with his first and second throws and hits the target with his third throw. State how much profit he has made. [1]
  - (ii)** Show that the probability that Mario's profit is \$0 is 0.184, correct to 3 significant figures. [2]
  - (iii)** Draw up a probability distribution table for Mario's profit. [3]
  - (iv)** Calculate his expected profit. [2]
- 7** In tests on a new type of light bulb it was found that the time they lasted followed a normal distribution with standard deviation 40.6 hours. 10% lasted longer than 5130 hours.
- (i)** Find the mean lifetime, giving your answer to the nearest hour. [3]
  - (ii)** Find the probability that a light bulb fails to last for 5000 hours. [3]
  - (iii)** A hospital buys 600 of these light bulbs. Using a suitable approximation, find the probability that fewer than 65 light bulbs will last longer than 5130 hours. [4]

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